management, a multilingual interface and, crucially, no dependence on a single vendor, the revamped system will be able to handle 125,000 tickets per minute as against the current rate of 25,000/min.

REBUILDING THE DIGITAL CORE

It is not just superfast ticketing the new system promises. It will also plug the major flaws in the current system that allowed loopholes to flourish and made revamping a long-standing necessity. For instance, a host of automated 'booking' apps—allegedly named 'Gadar', 'SpaceX' and 'Avenger'—were marketed on subscription to networks of agents who could commandeer ticket inventory before a genuine passenger could even open a browser. These tools would pre-load passenger details before the booking window opened, then fire off reservation requests the instant the clock struck Tatkal hour—bypassing IRCTC's captcha and bank OTP checks in seconds, even as a user was still typing in their name.

Worse, Aadhaar-authenticated IRCTC user IDs were being sold for Rs 360 each through over 40 Telegram and WhatsApp groups. In Mumbai, for example, in 2018, one individual reportedly controlled a network of more than 5,400 agents using software that evaded the IRCTC's captcha and automatically filled in passenger information. The operation allegedly earned him Rs 35 lakh per month. As many as half the login attempts in the first five minutes of Tatkal bookings came from bots. An IRCTC crackdown saw 25 million fake user IDs being blocked. The Railways is also using Aadhaar-linked authentication for e-ticket booking now. It is helping, but the new architecture will curb such nefarious misuse of the system.

The overhaul involves upgrading the entire network—from routers across every tier of the Railways' hierarchy to the bandwidth links connecting every terminal. Railway ticketing is organised in a decentralised, tiered architecture, engineers explain (see *How the New Ticketing System Works*). At the top sits the central data centre, which holds the master seat inventory. Beneath it are regional nodes, followed by station- and counter-level systems where a clerk enters passenger details and confirms a booking. At each layer, routers act like traffic controllers, sending booking requests up the hierarchy to the central servers and routing responses back to the terminal. The moment a passenger requests a seat, the system places it on hold for a few seconds—temporarily locking the record while the request travels between terminal and server—so no one else can claim it before the booking is confirmed and sent back. Each point in this chain is being given latest equipment and faster, reliable internet connections. Much of the work is already done: about 3,000 of the 5,788 routers at the zonal level have been replaced, 3,500 of the 5,000 booking terminals have migrated to the new application; the core software is nearing final testing.



ARUN KUMAR

Reservation Architecture

OLD VS NEW

How Indian Railways is rebuilding and improving its passenger reservation system

ARCHITECTURE

Old: Built in 1985 on a closed, proprietary architecture

New: Open-source, scalable architecture

DEPENDENCE

Old: Single-vendor; upgrades slow and expensive

New: No vendor lock-in; easier upgrades and expansion

BOOKING

Old: Capacity ceiling of 25,000 bookings per minute

New: Capacity raised to 125,000 per minute

TRAFFIC HANDLING

Old: No real-time traffic management; struggles during demand spikes

New: Real-time dynamic traffic management to handle demand surge

INFRASTRUCTURE

Old: Servers distributed across Delhi, Mumbai, Kolkata and Chennai

New: Mega data centre in Delhi with Disaster Recovery Centre in Hyderabad

SHIFTING PLATFORMS

The PRS has been running since 1985. Built to last perhaps a decade and a half, it has endured for nearly 40 years, making it among the world's oldest and most heavily used ticketing systems. Its software was written in Fortran 77, one of the earliest high-level programming languages. The new platform will run on open-source software: Red Hat's enterprise Linux environment and others. This means no single company will own the underlying code or control access. Any certified vendor can supply hardware for it, support it, or develop on top of it. If one manufacturer exits the market or becomes expensive, the Railways can switch without rebuilding from scratch, a bind it has long been trying to escape. This combination makes the five-fold jump in bookings per minute possible: unlike the old system, which could not be expanded just by adding more servers, the new one can scale horizontally on demand.